# 7.5 Properties of Trapezoids and Kites

Name _____ Period ____ Date _____

Geometry 7.5 · Student sheet

**I can…**

- find the angle measures of an isosceles trapezoid.
- use the Trapezoid Midsegment Theorem to find lengths.
- find the angle measures of a kite and name a quadrilateral as specifically as possible.

**Vocabulary** · fill in the missing word

**Trapezoid** — a _____ with exactly one pair of parallel sides (the bases); the other sides are the legs

**Isosceles trapezoid** — a trapezoid with _____ legs

**Base angles** — a pair of angles that _____ a base

**Midsegment of a trapezoid** — the segment that joins the _____ of the legs

**Kite** — a _____ with two pairs of consecutive congruent sides, but opposite sides not congruent

## Part 1 · Fold and cut

Fold a sheet in half and cut a shape that unfolds into an **isosceles trapezoid**. Then fold another sheet and cut a **kite**. Measure.

| Isosceles trapezoid | Measure |
|---|---|
| Base angles (bottom) | |
| Base angles (top) | |
| Diagonals | |

| Kite | Measure |
|---|---|
| The pair of angles that match | |
| The other two angles | |
| Angle where the diagonals cross | |

Two things that are always true about each shape:

## Part 2 · Notes: fill in the blanks

A **trapezoid** has exactly one pair of _____ sides (the bases).

In an isosceles trapezoid, each pair of base angles is _____, and the diagonals are _____.

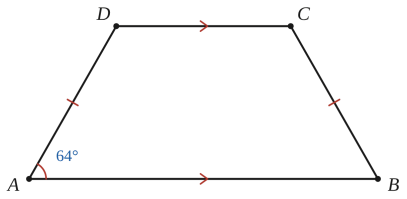
**Trapezoid Midsegment Theorem:** the midsegment is _____ to the bases, and its length is _____ the sum of the bases.

In a kite, the diagonals are _____, and exactly one pair of opposite angles is _____.

## Part 3 · Examples (work them with the class)

### Example 1

ABCD is an isosceles trapezoid with bases $\overline{AB}$ and $\overline{CD}$, and $m\angle A = 64°$. Find $m\angle B$, $m\angle C$, and $m\angle D$.




### Example 2

a. The bases of a trapezoid are 12 and 20. Find the length of the midsegment.

b. The midsegment of a trapezoid is 15 and one base is 11. Find the other base.

### Example 3

In kite EFGH, the angles between the congruent sides measure 70° and 50°. Find the measure of each of the other two angles.

### Example 4

Show that A(−2, 0), B(0, 3), C(4, 3), and D(6, 0) form a trapezoid. Is it isosceles?

## Part 4 · Partner practice

### Partner 1

The midsegment of a trapezoid is 14. The bases are $x + 4$ and $3x$. Find $x$ and both bases.

### Partner 2

In a kite, one of the congruent opposite angles measures 105° and one of the other angles measures 85°. Find the fourth angle.